

Anytime Coding for Gaussian Rate-Limited Noisy Feedback: Low Average Rates of Data Transmission

Aman Chawla

April 13, 2024

Abstract

In this paper, the authors present a result pertaining to the bounds on the error exponent of a continuous time gaussian channel with noisy feedback, when there is no limit on the feedback rate, other than capacity of the feedback link. Next, they study the noisy feedback reliability function in the case that there is only rate-limited noisy feedback available. They show how use of anytime coding in Decoder Phase 1 can be useful in this scenario.

1 Introduction

In [1] (see around Equation (3.4)) we studied the decoder phase 1 feedback. We assumed that the feedback rate was $\frac{\log(M)}{T_3}$ bits per second where M is the number of messages and T_3 is the phase 1 decoder feedback signaling duration. Suppose, however, we allow only one among $\sqrt{M}$ number of messages out of the total of M messages to be sent during T_3 seconds, that is, we halve the rate allowed. This may have several implications, but in this work we consider the direct impact on the error exponent calculation.

This paper is organized as follows. In Section 2 we present a negative result prior to the modification suggested above. In Section 3 we consider rate-limited feedback as suggested above, under the usual feedback coding choice, namely, the identification-oriented scheme and the orthogonal signaling scheme. In Section 4 we consider rate-limited feedback under a scheme which replaces these two schemes with an anytime code. We conclude with a discussion in Section 5.

2 The Perfect Feedback Exponent is Not Achieved Everywhere by Every Asymptoting Two Phase Noisy Feedback Scheme

In the SM thesis [1], a simple two phase scheme is presented for the noisy gaussian feedback setting in continuous time. The achieved reliability function is lower bounded there. Here we make use of a bound from [2, 3]. It is an upper bound on the perfect feedback reliability function.

Figure 1 shows both the bounds plotted on the same graph. for different values of the feedback capacity (increasing from top to bottom, with perfect feedback in the bottom-most subplot) and a fixed, large value of the Cprime parameter of the forward channel. Figure 2 shows the same situation but with a small value of the Cprime parameter of the forward channel.

The figures suggest that, depending upon the value of Cprime, the rudimentary two phase scheme of [1] performs optimally only for low rates. What makes this come about is that the pre-factor multiplying $(1 - \frac{r}{C_{forward}})$ is $(\frac{1}{4C_{forward}} + \frac{1}{C_{feedback}})^{-1}$. The first term in the inverse sum ensures that $C_{feedback}$ cannot have an unconstrained effect.

3 Square-root Feedback: Usual Scheme

Consider the allowable use of the noisy feedback channel to be limited to half of the rate allowed in Section 2. Thus we only allow $R' = \frac{\log \sqrt{M}}{T_3} = \frac{\log M}{2T_3}$ bits per second in Decoder phase 1 feedback. If we then use orthogonal signaling, since we're operating at half the rate, we'll see the half-rate error exponent value from Gallager's error exponent formula. If we use an identification oriented code, since $R_{dID} = \frac{K \ln \ln M}{T_3}$, our new identification rate will be $R'_{dID} = \frac{K \ln \ln \sqrt{M}}{T_3} = R_{dID} - \frac{K \ln 2}{T_3}$. We could do either of these implementations, but there would be no novelty in the approach; these are standard codes. For the sake of comparison with the modified scheme to be introduced next, it will therefore suffice to work at the level of the final error exponent of the overall scheme, Equation (3.25) of [1].

4 Square-root Feedback: Modified Scheme

In this section we consider what happens when the rate is limited as in the previous section, Section 3, but the decoder phase one feedback is performed using an anytime code. Let W' be the delay of the phase 1 decoder anytime feedback. The rate is as in the previous section, R' . During the period $D = W'(T_1 + T_4 + T_2)$ the exponent is

$$P_e^{(3)} \leq K'' \exp(-W' E_{orth}(R') T_3). \quad (1)$$

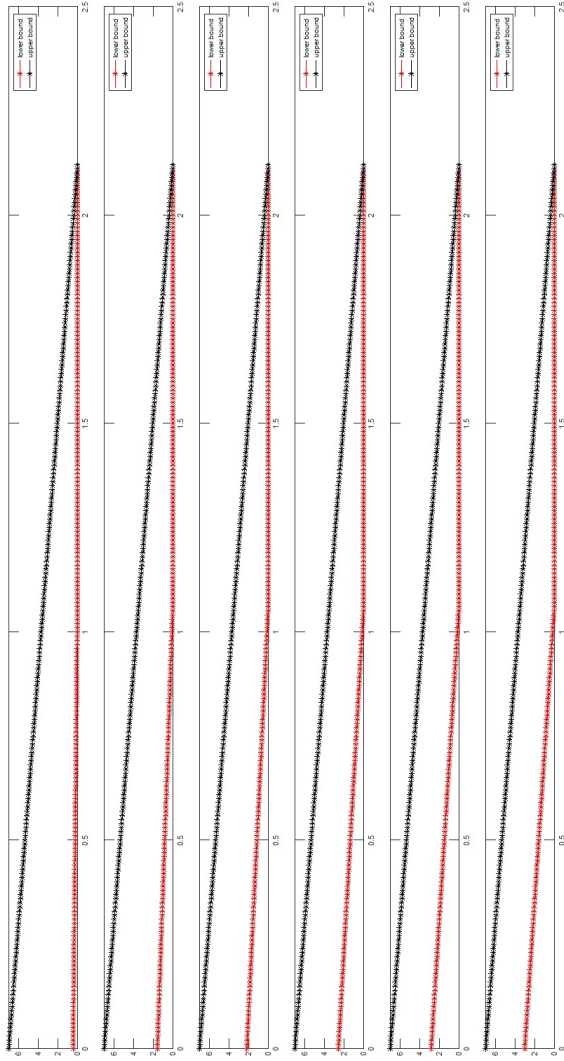


Figure 1: High value of C_{prime} . The horizontal axis of each sub-plot is the average transmission rate, and the vertical axes represent the error exponent bounds. The forward channel capacity is taken as 1.06 bits per second.

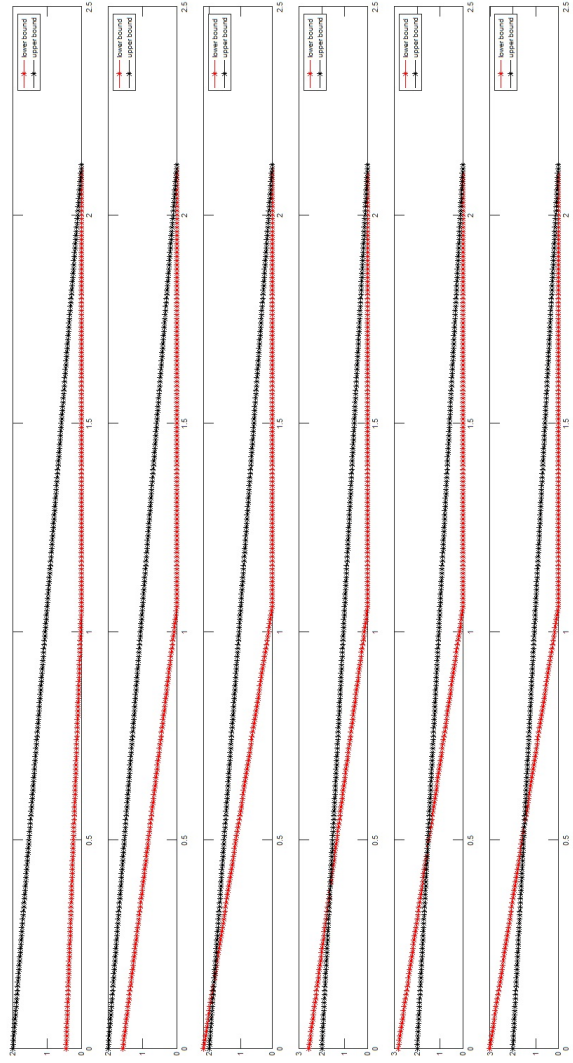


Figure 2: Low value of C_{prime} . The horizontal axes are the average transmission rate and the vertical axes represent the error exponent bounds. Where the lower bound (red) crosses above the upper bound (black), the upper bound indicates the exponent achieved. The forward channel capacity is taken as 1.06 bits per second.

The probabilities (only the exponents are given) of the three types of error are: (i) $4C_1T_2$, (ii) $W'E_{orth}(R')T_3$ and (iii) $WE_{orth}(R'')T_4$.

The expected decoding time will increase by a factor of $(1+pWW')$ where p is the sum of a negligible value and $1e-5$. Using the error probability computation and expected decoding time computation, and following the steps outlined in [1] we find that the overall scheme's error exponent has a lower bound:

$$\left(\frac{1}{W'E_{orth}(R')} + \frac{1}{4C_1}\right)^{-1} \left[\frac{1}{1+pWW'} - \frac{\bar{R}}{C_1}\right]. \quad (2)$$

When the average rate considered is $\bar{R} = \frac{C_1}{1+pWW'}$, this lower bound zeroes out. Using some analytical work and some help from the MATLAB programming environment, we find that the lower bound above beats Equation (3.25) of [1] when W' is made suitably large. Thus the anytime delay of decoder phase one feedback directly helps in beating the bound proposed by Chawla in 2006, in the low rate regime.

5 Conclusion

This paper highlights the intriguing nature of noisy feedback. We may conclude that for rate-limited noisy feedback, for a wide range of parameter choices, using an anytime code in decoder phase one feedback allows one to have an error exponent larger than that of the original scheme for low average rates of data transmission.

References

- [1] Aman Chawla. Reliability of a gaussian channel in the presence of gaussian feedback. *S.M. Thesis*, 2006.
- [2] Peter Berlin, Barış Nakiboğlu, Bixio Rimoldi, and Emre Telatar. A simple converse of burnashev's reliability function. *Information Theory, IEEE Transactions on*, 55(7):3074–3080, 2009.
- [3] Aman Chawla and Salvatore Domenic Morgera. On the continuous time additive gaussian noise channel in the presence of perfect feedback. In *2020 IEEE Information Theory Workshop (ITW)*, pages 1–5. IEEE, 2021.